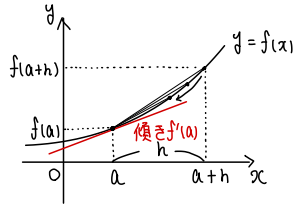
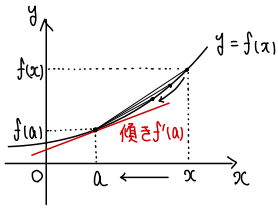


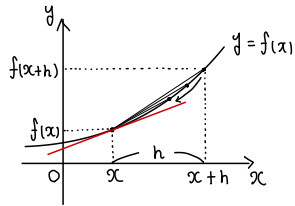
微分係数と導関数

微分係数 $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$



微分係数 $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

導関数 $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$



(例) 次の関数を導関数の定義に従って微分せよ。

(1) $f(x) = \frac{1}{x}$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h(x+h)x} \\ &= \lim_{h \rightarrow 0} \left(-\frac{1}{(x+h)x} \right) \\ &= -\frac{1}{x^2} \end{aligned}$$

(2) $f(x) = \sqrt{x}$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$